

1. Planificación de Segmentación (Subnetting)

Se ha implementado un esquema de direccionamiento basado en el bloque 172.16.0.0 con una máscara de 19 bits (/19). Este diseño permite la separación lógica de los departamentos críticos de la clínica en ambas sucursales.

Segmento	ID de Red	Máscara	Rango Asignable	Gateway	Uso / VLAN
DC Sucursal 1	172.16.0.0	255.255.224.0	172.16.0.2 - 172.16.31.254	172.16.0.1	Servidores Core S1
RRHH S1	172.16.32.0	255.255.224.0	172.16.32.2 - 172.16.63.254	172.16.32.1	Gestión Adm S1
Salud S1	172.16.64.0	255.255.224.0	172.16.64.2 - 172.16.95.254	172.16.64.1	Puestos Médicos S1
WiFi S1	172.16.96.0	255.255.224.0	172.16.96.2 - 172.16.127.254	172.16.96.1	Acceso Dinámico
DC Sucursal 2	172.16.128.0	255.255.224.0	172.16.128.2 - 172.16.159.254	172.16.128.1	Servidores Core S2
RRHH S2	172.16.160.0	255.255.224.0	172.16.160.2 - 172.16.191.254	172.16.160.1	Gestión Adm S2
Salud S2	172.16.192.0	255.255.224.0	172.16.192.2 - 172.16.223.254	172.16.192.1	Puestos Médicos S2

Segmento	ID de Red	Máscara	Rango Asignable	Gateway	Uso / VLAN
WiFi S2	172.16.224.0	255.255.224.0	172.16.224.2 - 172.16.255.254	172.16.224.1	Acceso Dinámico

2. Configuración de Servidores de Red

Los servicios se encuentran distribuidos para garantizar alta disponibilidad. Se detallan a continuación las direcciones IP estáticas de los servidores principales.

Servicio	IP Sucursal 1	IP Sucursal 2	Descripción del Rol
Mensajería SMTP	172.16.0.2	172.16.128.2	Correo Interno clinica.org
Archivos FTP	172.16.0.3	172.16.128.3	Repositorio de Datos
Nombres DNS	172.16.0.4	172.16.128.4	Resolución de Hostnames
Web Portal	172.16.0.5	172.16.128.5	Site Institucional
Sistema Médico	172.16.0.6	172.16.128.6	Panel para Profesionales
Direccionamiento DHCP	172.16.0.7	172.16.128.7	Gestión de IPs WiFi

3. Seguridad y Acceso de Usuarios

Definición de cuentas para los servicios de correo y transferencia de archivos.

- **Administración (RRHH):** Cuentas rrhh1 a rrhh4. Poseen permisos totales (Lectura, Escritura, Eliminación).

- **Staff Médico:** Cuentas medico1 a medico4. Restringidos a Solo Lectura y Listado.
- **Acceso WiFi:** Cuentas wifi1 a wifi4. Permisos de Lectura y Escritura.

4. Conectividad Inalámbrica (Access Points)

La red inalámbrica utiliza autenticación robusta para todos los dispositivos móviles.

Parámetro	Configuración
SSID Sucursal 1	Clinica-WiFi-Suc1
SSID Sucursal 2	Clinica-WiFi-Suc2
Método de Seguridad	WPA2-PSK
Frase de Acceso	clinica2025

5. Verificación de Funcionamiento

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
✓	Successful	PC3...	PC1-RRHH...	ICMP	Green	0.000	N	0	(e...)	(delete)
✓	Successful	PC3...	PC2-RRHH...	ICMP	Red	0.000	N	1	(e...)	(delete)
✓	Successful	PC3...	PC2-RRHH...	ICMP	Orange	0.000	N	2	(e...)	(delete)

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 172.16.64.2

Pinging 172.16.64.2 with 32 bytes of data:

Reply from 172.16.64.2: bytes=32 time=10ms TTL=128

Reply from 172.16.64.2: bytes=32 time=1ms TTL=128

Reply from 172.16.64.2: bytes=32 time=1ms TTL=128

Reply from 172.16.64.2: bytes=32 time=1ms TTL=128

Ping statistics for 172.16.64.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 10ms, Average = 3ms

Cisco Packet Tracer PC Command Line 1.0

C:\>ping web-clinica.org

Pinging 172.16.128.5 with 32 bytes of data:

Request timed out.

Reply from 172.16.128.5: bytes=32 time<1ms TTL=127

Reply from 172.16.128.5: bytes=32 time=9ms TTL=127

Reply from 172.16.128.5: bytes=32 time<1ms TTL=127

Ping statistics for 172.16.128.5:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

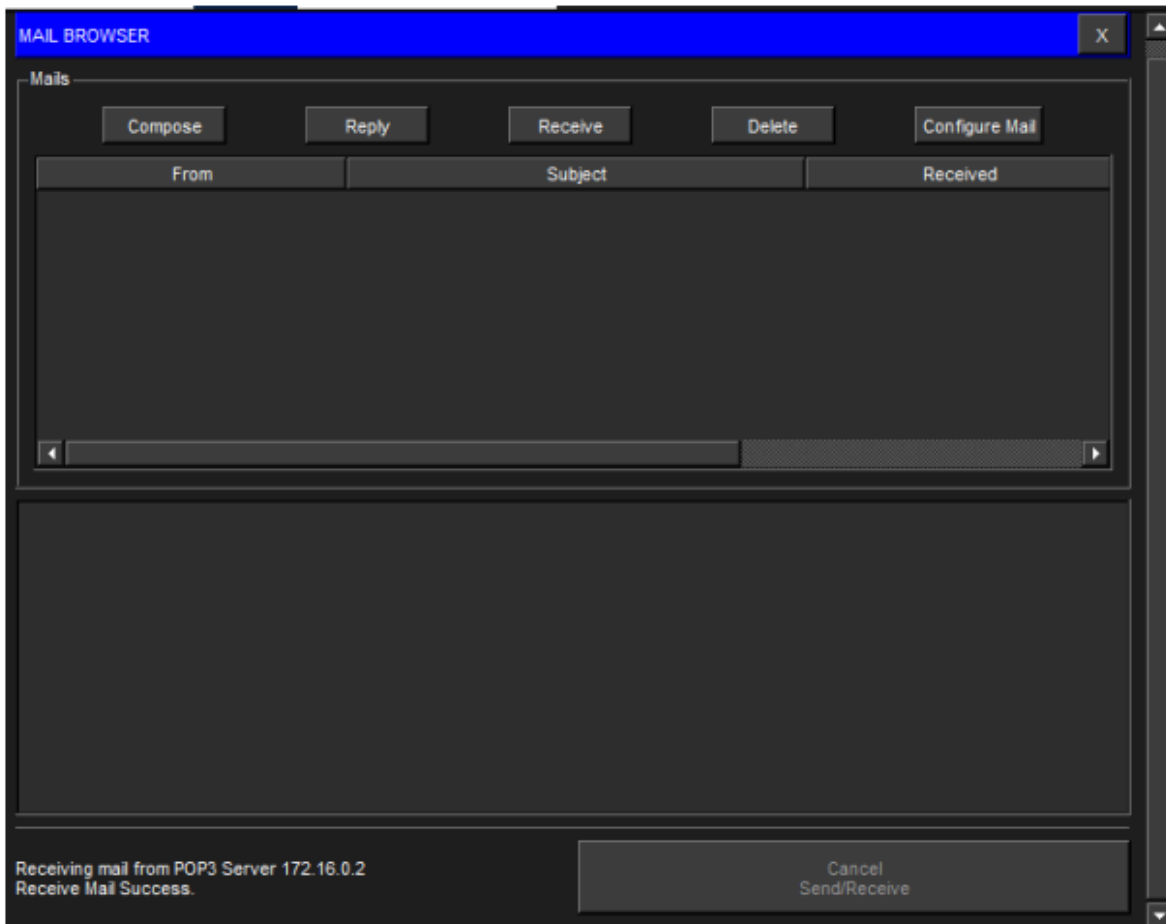
Minimum = 0ms, Maximum = 9ms, Average = 3ms

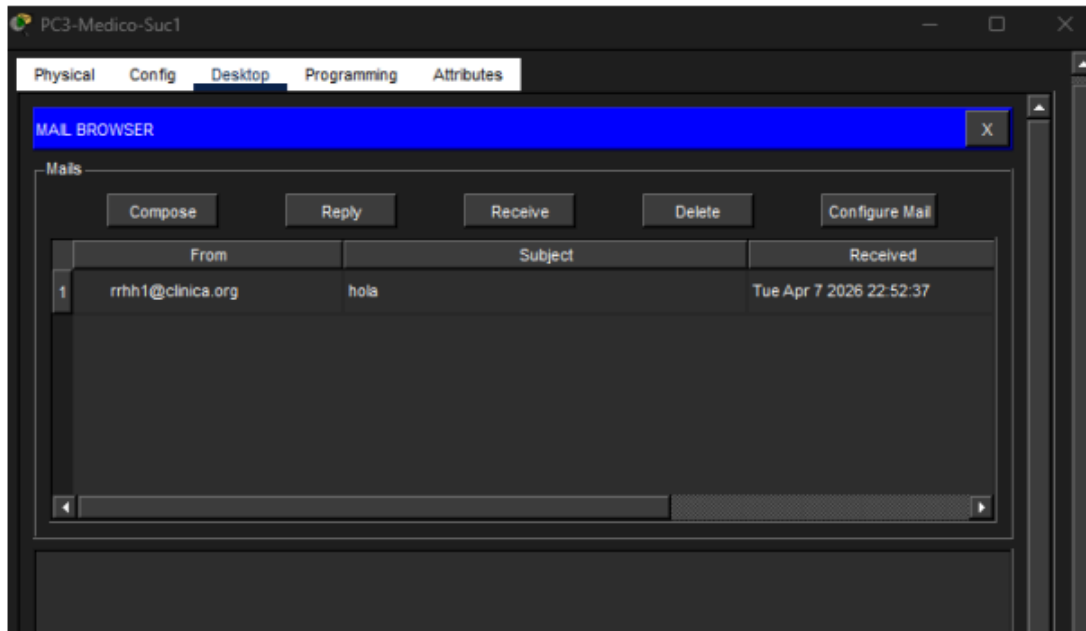
```
C:\>tracert 172.16.128.2
```

```
Tracing route to 172.16.128.2 over a maximum of 30 hops:
```

1	0 ms	0 ms	0 ms	172.16.64.1
2	0 ms	0 ms	0 ms	10.255.0.2
3	*	0 ms	0 ms	172.16.128.2

```
Trace complete.
```





```
C:\>ftp 172.16.0.3
Trying to connect...172.16.0.3
Connected to 172.16.0.3
220- Welcome to FT Ftp server
Username:rrhh1
331- Username ok, need password
Password:*****

!Error ftp://172.16.0.3/ (No such Account)
332- Need account for login
```

```

C:\>ftp
C:\>ftp 172.16.0.3
Trying to connect...172.16.0.3
Connected to 172.16.0.3
220- Welcome to PT Ftp server
Username:rrhh1
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 172.16.0.3:
0   : asa842-k8.bin                               5571584
1   : asa923-k8.bin                               30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin      33591768
3   : c1841-ipbase-mz.123-14.T7.bin                13832032
4   : c1841-ipbasek9-mz.124-12.bin                 16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin       33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin      33591768
7   : c2600-i-mz.122-28.bin                        5571584
8   : c2600-ipbasek9-mz.124-8.bin                  13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin     50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin      33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin              5571584
12  : c2800nm-ipbasek9-mz.124-8.bin                 15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin       33591768
14  : c2950-i6q412-mz.121-22.EA4.bin              3058048
15  : c2950-i6q412-mz.121-22.EA8.bin              3117390
16  : c2960-lanbase-mz.122-25.FX.bin               4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin             4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin             4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin      8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin       10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin          33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin         83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin        505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG           159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG           184530138
26  : ie9k_iosxe.17.09.04.SPA.bin                  596133776
27  : ir800-universalk9-bundle.SPA.156-3.M.bin      160968869
28  : ir800-universalk9-mz.SPA.155-3.M              61750062
29  : ir800-universalk9-mz.SPA.156-3.M              63753767
30  : ir800_yocto-1.7.2.tar                         2877440
31  : ir800_yocto-1.7.2_python-2.7.3.tar           6912000
32  : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33  : pt1000-i-mz.122-28.bin                        5571584
34  : pt3000-i6q412-mz.121-22.EA4.bin              3117390

```

